

Next Generation Arithmetic for Edge Computing

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Abstract—Arithmetic is a key component and is ubiquitous in today's digital world, ranging from embedded to high-performance computing systems. With machine learning at the fore in a wide range of application domains from wearables to automotive to avionics to weather prediction, sufficiently accurate yet low-cost arithmetic is the need for the day. Recently, there have been several advances in the domain of computer arithmetic, which includes high-precision anchored numbers from ARM, posit arithmetic, bfloat16, etc. as an alternative to IEEE 754-2008 compliant arithmetic. Optimizations on fixed-point and integer arithmetic are also being pursued actively for low-power computing architectures. Furthermore, approximate computing and transprecision/mixed-precision computing have been exciting areas of research forever. While academic research in the domain of computer arithmetic has a long history, industrial adoption of some of these new data types and techniques is in its early stages and expected to increase in the future. bfloat16 is an excellent example for this. In this paper, we bring academia and industry together to discuss the latest results and future directions for research in the domain of next-generation computer arithmetic, especially for edge computing.

Index Terms—Computer arithmetic, floating point arithmetic, mixed-precision arithmetic, IEEE 754-2008 standard

I. INTRODUCTION

Computer arithmetic is ubiquitous in applications ranging from an embedded domain such as smartphones to high-performance computing (HPC) applications like weather modeling. Specifically, in embedded systems, since the platforms have performance limitations due to limited power and area budgets, using appropriate arithmetic is desirable. The constraints on power and area footprints combined with demand for high performance in applications like edge computing in internet-of-things (IoT) have created tremendous challenges for researchers and computer architects [1]. Over the years in response to this challenge, researchers have developed hardware-efficient implementations of computer arithmetic [2]. Recently, there have been attempts to design new representations to replace the existing ones [3].

Major semiconductor manufacturers involved in efficient domain-specific accelerator designs are investigating new arithmetic formats for their applications of interest [4] [5]. For example, the recently proposed *bfloat16* format supports

similar dynamic range as single-precision IEEE 754 compliant floating point number, but with lower precision. The *bfloat16* format is supported in several products by Intel, Google, and ARM. Similarly, several academic researchers have designed and developed software and hardware tools that can automatically generate efficient arithmetic hardware. The FloPoCo project is one such example. In this paper, we describe and review the following complementing approaches that lead to an optimum computer arithmetic for edge computing:

- Firstly, we describe an application-specific arithmetic where the focus is on operator transformation and computing exactly required arithmetic.
- Secondly, we describe an arithmetic architecture and approaches for efficiently implementing them on field programmable gate arrays (FPGAs) mainly targeting artificial intelligence (AI) applications.
- Thirdly, we describe a hardware-oriented approximate computing for speech recognition and keyword spotting.
- Finally, we discuss a new datatype called *posit arithmetic* that is proposed as a drop-in replacement for IEEE 754 format and arithmetic.

The rest of the paper is organized as follows. In Section II, we discuss application-specific arithmetic. FPGA implementation and support for arithmetic is discussed in Section III. Applications of approximate computing are described in section IV. Posit arithmetic and its hardware cost are discussed in Section V as an example of alternate number formats. Finally, we conclude our work in Section VI.

II. APPLICATION-SPECIFIC ARITHMETIC DESIGN

Conventional arithmetic units designed for general-purpose processors have to be most generally useful. A good illustration is how the fused multiply-and-add became the floating-point unit of choice at the turn of the century: it could replace an adder and a multiplier, but also enable efficient and flexible implementations of division, square root, elementary functions, etc. [6], while improving the performance on linear algebra computations (the measure of all things in high-performance computing). This particular technical choice is now being reconsidered for energy-efficiency reason, but it remains that